

Marcus DeVries

Boone, CO | marcusdevries59@gmail.com | (719) 369-9668

LinkedIn: www.linkedin.com/in/marcus-devries-77847570 | US Person - ITAR/EAR Compliant

Robotics & AI Engineer

Expert in developing autonomous systems that integrate AI intelligence with electromechanical precision. 15+ years of experience in robotics R&D, precision manufacturing, and full-stack prototyping, supported by advanced degrees in Machine Learning and Computer Science. Proven ability to lead R&D projects from concept to functional prototype, specializing in machine learning applications for real-world control, diagnostics, and autonomous operation.

Technical Competencies

AI & Machine Learning: Supervised & Unsupervised Learning, TensorFlow, PyTorch, Predictive Maintenance, Computer Vision, Natural Language Processing (NLP), Generative Adversarial Networks (GANs), Model Training & Deployment

Robotics & Autonomy: Sensor Integration (LIDAR, 3D Scanning), Actuator Control, System Architecture, Autonomous Workflow Design, ROS Concepts, Control Systems (PID, Kinematics, Motion Planning)

Programming & Software: Python, C++, Java, G-Code, Siemens NX, SolidWorks, Fusion 360, Git, Arduino

Development & Operations: Software Development Life Cycle (SDLC), Agile & Waterfall Methodologies, Modular System Design, Containerization

Electromechanical Systems: Mechatronic Prototyping, CNC Machinery, Precision Actuation, GD&T, System Diagnostics & Troubleshooting, Circuit Design & PCB Layout

Engineering Process: R&D Lifecycle, Technical Documentation, Cross-Functional Collaboration, Testing & Validation (V&V)

Professional Experience

Ascended Technologies | Boone, CO

Founder & Principal Robotics Engineer | 2021 – Present

- Spearheading R&D for an AI-driven, autonomous CAD/CAM software agent, managing development through Agile SDLC practices from concept to active prototyping.
- Architected and currently developing a modular machine learning framework using Python and TensorFlow/PyTorch, applying both supervised and unsupervised learning models to optimize manufacturing parameters for closed-loop production.
- Developing a containerized computer vision diagnostic agent using convolutional neural networks (CNNs) for early anomaly detection; built a scalable, modular architecture to facilitate training and deployment.
- Designed the full system architecture for a mechatronic animatronic platform, integrating LIDAR for navigation and defining sensor-to-actuator control loops for reactive behavior.

A Plus Welding | Avondale, CO

CAM Programmer & Systems Integrator | 2020 – 2021

- Programmed and optimized complex CNC toolpaths, reducing production time by 20% through algorithmic parameter optimization.
- Engineered custom fixtures and jigs to improve system repeatability and operational efficiency.

Formacore | Fountain, CO

Systems Engineering Lead | 2018 – 2019

- Developed and programmed advanced manufacturing systems for aerospace and satellite components.
- Integrated Faro 3D scanning as a closed-loop inspection system, overlaying point cloud data with CAD models to automatically verify part conformance and dimensional accuracy.
- Trained technical teams on system operation and quality assurance protocols.

United Launch Alliance | Pueblo, CO

Precision Manufacturing & Technical Support Engineer | 2015 – 2018

- Programmed and validated CNC systems for high-tolerance aerospace flight hardware, ensuring 100% conformance to rigorous engineering specifications.
- Collaborated with design engineers to ensure manufacturability and system reliability for critical launch components.

Selected Projects & Research

- AI-Powered Autonomous Manufacturing Agent: Developing a full-stack software agent that automates the entire manufacturing planning process, from CAD input to machine-ready code, representing a significant step towards factory autonomy.
- Medical Image Diagnostic Agent: Architected and trained a deep learning model (CNN-based) in TensorFlow/PyTorch for the early detection of pathologies, managing the project from data curation and model development to validation documentation.
- Wildlife Population Analytics SaaS (Concept): Architected a drone-based SaaS platform concept utilizing real-time video feeds and computer vision to track and monitor animal populations for conservation agencies, extending my work in AI diagnostics and autonomous systems.
- Animatronic Robotic System ("Facehugger"): Designed the full system architecture for a mechatronic animatronic platform, integrating LIDAR for navigation and defining sensor-to-actuator control loops for reactive behavior.
- PCB Environmental Control Systems: Designed and implemented printed circuit boards (PCBs) for lighting and control systems in hazardous environments. Interpreted electrical schematics and executed board layouts to ensure system reliability and safety compliance.
- NASA Rover Challenge (COSGC): Contributed to the design and build of a competitive Martian rover prototype, gaining hands-on experience in collaborative robotics engineering within a structured team.

Education

M.S., Artificial Intelligence & Machine Learning | Colorado State University Global

Relevant Coursework: Neural Networks, Natural Language Processing, Computer Vision

Thesis/Capstone: Developed a goal-oriented chatbot using a BERT-based architecture for health metric tracking.

B.A.S., Computer Science | Colorado State University Global

Core Focus: Software Engineering, C++, Java, Data Structures & Algorithms

A.A.S., Machining Technology | Pueblo Community College (Dean's List, Phi Theta Kappa)

A.A.S., Welding Technology | Pueblo Community College (Dean's List, Phi Theta Kappa)